Yuanchao Xu

xu.yuanchao.3a@kyoto-u.ac.jp

Room 4-405, Maskawa Building for Education and Research, Kyoto University

October 10, 2026

Working Experience

Kyoto University
Program-Specific Researcher, Department of Mathematics
October 2026 - August 2030

Kyoto University
Program-Specific Researcher, SACRA
September 2025 - September 2026

University of Alberta
Visiting Postdoc, Applied Mathematics
April 2026 - August 2026

Education Experience

University of Alberta
Doctor of Philosophy (*Dissertation Award*), Applied Mathematics
September 2021 - August 2025

University of Manitoba
Master of Science, Mathematics
September 2018 - August 2020

University of Oregon
Bachelor of Science (*Cum Laude*), Mathematics
January 2015 - December 2017

Research Interest

My research interest lies in data-driven methods for stochastic dynamical system. In particular, I am interested in building up theoretical framework for learning stochastic Koopman operator and applying it into complex real-world problems, e.g., computational neuroscience, reinforcement learning and generative modeling, etc.

Research Submitted / In Preparation

Talk

1. *Generative Modeling through Koopman Spectral Analysis*
   Massachusetts Institute of Technology, Cambridge, MA, April 27, 2026
2. *Generative Modeling through Koopman Spectral Analysis*
   CMS Special Seminar
   California Institute of Technology, Pasadena, CA, April 13, 2026
3. *Generative Modeling through Koopman Spectral Analysis*
   Probability Seminar
   University of Oregon, Eugene, OR, April 8, 2026
4. *Generative Modeling through Koopman Spectral Analysis*
   MLDS 5 & DEDS 2026
   Kyoto University, Kyoto, Japan, February 9-13, 2026
5. *Generative Modeling through Koopman Spectral Analysis*
   Machine Learning and Dynamical Systems Seminar
   Alan Turing Institute, London, UK, January 8, 2026 (Online)
6. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
   Fudan University, Shanghai, China, January 5, 2026
7. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
   European Molecular Biology Laboratory, Rome, Italy, November 24, 2025 (Online)

8. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
   Laboratoire Hubert Curien, Saint-Étienne, France, October 14, 2025
9. *Perturbation method for learning stochastic Koopman operator*
   CREST
   RIKEN, Kobe, Japan, December 26-28, 2024
10. *Extracting Dynamics from Complex Systems: Deterministic and Stochastic Perspectives*
    East China University of Science and Technology, Shanghai, China, December 19, 2024

Visiting

1. Massachusetts Institute of Technology, Cambridge, MA, April 27-30
2. Cornell University, Ithaca, NY, April 16-18
3. California Institute of Technology, Pasadena, CA, April 13-15
4. University of California, Santa Barbara, CA, April 10th
5. University of Oregon, Eugene, OR, April 6-9
6. Fudan University, Shanghai, China, January 5-9, 2026
7. Laboratoire Hubert Curien, Saint-Étienne, France, October 14-15, 2025
8. European Molecular Biology Laboratory, Rome, Italy, October 11-13, 2025
9. Istituto Italiano di Tecnologia, Genoa, Italy, October 9-10, 2025
10. Ehime University, Matsuyama, Japan, December 23-25, 2024

Reviewer

ICML, ICLR, AISTATS, NeurIPS, IEEE-TNNLS, IEEE-TETCI, Physica D, Applied Mathematical Modelling

Service

Young Associate Editor: DCDS-I

Awards

- Faculty of Science Doctoral Dissertation Award, 2025
- Mary Louise Imrie Graduate Student Award, 2025
- Dr. Josephine M. Mitchell Recruitment Scholarship, 2021
- International Graduate Student Entrance Scholarship, 2020
- International Graduate Student Entrance Scholarship, 2018
- Clarence and Lucille Dunbar Scholarship, 2017

Teaching Experience

Math 102(lab) Applied Linear Algebra	*Winter 2025*
Math 209(lab) Calculus for Engineering 3	*Fall 2024*
Math 209(lab) Calculus for Engineering 3	*Spring 2024*
Math 102(lab) Applied Linear Algebra	*Winter 2024*
Math 201(lab) Differential Equations	*Fall 2023*
Math 101(lab) Calculus for Engineering 2	*Spring 2023*
Math 201(lab) Differential Equations	*Winter 2023*
Graduate Research Assistant	*September 2020 - August 2021*
Graduate Teaching Assistant	*September 2018 - August 2020*
Undergraduate Teaching Assistant	*September 2016 - December 2017*
Undergraduate Math Tutor	*April 2016 - December 2017*